

Robustness of the EFC Growth-Sector Signal

Leave-One-Out, Sound-Horizon, and Amplitude-Prior
Controls for the Hubble-Friction Channel

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Abstract. Energy-Flow Cosmology (EFC) modifies the Friedmann equation through an entropy-gradient coupling term governed by a single late-time parameter α . A joint Bayesian fit to 14 BAO distance measurements and 7 $f\sigma_8$ growth-rate data points yields $\alpha = -1.00 \pm 0.46$ (2.20σ) under controlled nuisance priors (N2a mode: $r_d = 147.09$ Mpc, $\sigma_8 \sim \mathcal{N}(0.81, 0.02)$). This note subjects the signal to three independent robustness controls: (N1) sound-horizon independence, (N2) amplitude-prior sensitivity, and (T7) leave-one-out (LOO) removal of each $f\sigma_8$ data point. Result: all 7 LOO runs pass ($|\alpha|/\sigma \geq 1.84$ and $\Delta\text{AIC} \leq 0$), the signal range is narrow ($\alpha \in [-1.11, -0.88]$), and parameter correlations are stable. The $\sim 2\sigma$ hint for late-time growth suppression is not an artefact of calibration choices, prior widths, or any single measurement. It is a distributed, robust, sub-detection-threshold signal awaiting confirmation or falsification by forthcoming surveys.

1. Introduction

The Λ CDM model fits current cosmological data with six parameters but relies on a cosmological constant Λ whose physical origin remains unexplained. Energy-Flow Cosmology (EFC) proposes that late-time cosmic acceleration arises from entropy-gradient coupling in the cosmological medium, parameterised at the background level by a single coupling constant α that modifies the Friedmann equation:

$$E^2(a) = E_{\Lambda\text{CDM}}^2(a) [1 + \alpha T(a)], \quad (1)$$

where $T(a)$ is a gate function that activates at late times ($z \lesssim 2$) and vanishes in the early universe, ensuring automatic recovery of Λ CDM at high redshift [1].

A joint fit to BAO and $f\sigma_8$ data in Phase 2.2 of the EFC validation programme found $\alpha = -1.00 \pm 0.46$, corresponding to a 2.20σ preference for $\alpha < 0$ (late-time growth suppression) under the N2a nuisance configuration [2]. The question addressed here is whether this signal is robust—or whether it is an artefact of the sound-horizon calibration, the σ_8 prior width, or a single anomalous data point.

2. Method

2.1 Data

The analysis uses two compilations:

- **BAO (14 points):** D_M/r_d and D_H/r_d measurements from 6dFGS, SDSS MGS, BOSS DR12 (3 redshift bins), eBOSS LRG/QSO/ELG, and Ly- α cross-correlation, spanning $z = 0.106\text{--}2.334$. Full 6×6 covariance from the BOSS consensus release is used where available.
- **Growth rate (7 points):** $f\sigma_8(z)$ measurements from 6dFGS ($z = 0.02$), SDSS MGS ($z = 0.15$), BOSS DR12 ($z = 0.38, 0.51, 0.61$), VIPERS ($z = 0.77$), and FastSound ($z = 0.85$).

2.2 Pipeline: MVP-G1 Hubble Friction Channel

The EFC modification enters through the Hubble rate $H(a)$ via Eq. (1). The growth-rate response $f\sigma_8(z)$ is computed by solving the standard growth ODE with the modified $H(a)$. The Poisson equation is unmodified ($\mu = 1$); the sole new degree of freedom is α .

2.3 Nuisance configuration (N2a mode)

To isolate the α signal from known degeneracies:

- Sound horizon fixed: $r_d = 147.09$ Mpc (Planck 2018 best fit).
- Matter density: $\Omega_m \sim \mathcal{N}(0.315, 0.007)$ (Planck prior).
- Growth amplitude: $\sigma_8 \sim \mathcal{N}(0.81, 0.02)$ (tight prior).
- Hubble constant: $H_0 \sim \mathcal{N}(67.4, 0.5)$ km s⁻¹ Mpc⁻¹.

2.4 Three robustness controls

- N1 Sound-horizon independence:** Vary r_d across its full Planck posterior range. If α is stable, the signal is not a calibration artefact.
- N2 Amplitude-prior control:** Compare α under flat σ_8 prior vs. tight $\sigma_8 \sim \mathcal{N}(0.81, 0.02)$. If α strengthens under the tight prior, the signal is not prior-driven.
- T7 Leave-one-out (LOO):** Remove each of the 7 $f\sigma_8$ data points in turn and refit. If α remains significant in all 7 runs, no single measurement drives the result. Pass criterion: $|\alpha|/\sigma \geq 1.7$ AND $\Delta\text{AIC} \leq 0$ in every run.

3. Results

3.1 N1: Sound-horizon independence

The α estimate is insensitive to r_d within its Planck posterior range. Shifting r_d by $\pm 1\sigma$ produces $|\Delta\alpha| < 0.05$, negligible compared to the statistical uncertainty of ± 0.46 . The signal is not a sound-horizon calibration artefact.

3.2 N2: Amplitude-prior control

Under the tight prior $\sigma_8 \sim \mathcal{N}(0.81, 0.02)$, the signal *strengthens* from $\sim 1.7\sigma$ (flat prior) to 2.20σ . This is expected: the α – σ_8 anti-correlation ($r \approx -0.27$) means a tighter σ_8 reduces the degeneracy and sharpens the α constraint. The signal is not a prior-width artefact.

3.3 T7: Leave-one-out robustness

Table 1: Leave-one-out results (N2a mode). Reference: $\alpha = -1.00 \pm 0.46$ (2.20σ , $\Delta\text{AIC} = -2.91$).

Drop	z	$f\sigma_8 \pm \sigma$	$\alpha \pm \sigma$	$ \alpha /\sigma$	ΔAIC	ΔBIC	$r(\alpha, \Omega_m)$	Pass
[0]	0.02	0.360 ± 0.040	-0.88 ± 0.48	1.84	−1.59	−0.59	+0.622	✓
[1]	0.15	0.490 ± 0.055	-1.11 ± 0.46	2.42	−3.97	−2.97	+0.598	✓
[2]	0.38	0.430 ± 0.054	-0.98 ± 0.46	2.12	−2.58	−1.58	+0.586	✓
[3]	0.51	0.452 ± 0.057	-1.03 ± 0.47	2.22	−2.97	−1.97	+0.588	✓
[4]	0.61	0.457 ± 0.052	-1.04 ± 0.48	2.19	−3.20	−2.21	+0.608	✓
[5]	0.77	0.420 ± 0.060	-1.00 ± 0.47	2.12	−2.70	−1.71	+0.583	✓
[6]	0.85	0.380 ± 0.080	-0.97 ± 0.46	2.09	−2.54	−1.54	+0.594	✓

Robustness score: 7/7 (100%). All LOO runs satisfy the pass criterion ($|\alpha|/\sigma \geq 1.7$ and $\Delta\text{AIC} \leq 0$).

Key observations:

- The α range across all LOO runs is $[-1.11, -0.88]$ —a spread of only 0.23, far smaller than the statistical uncertainty.
- The most influential point is $z = 0.02$ (6dFGS, local universe): dropping it reduces significance to 1.84σ but well above the 1.7σ threshold.
- Dropping $z = 0.15$ (SDSS MGS) *increases* significance to 2.42σ ; this point mildly pulls α toward zero.
- Correlations are stable: $r(\alpha, \Omega_m) \approx +0.59$ and $r(\alpha, \sigma_8) \approx -0.27$ in all runs.

4. Discussion

4.1 Summary of controls

Table 2: MVP-G1 robustness control summary.

Control	Result	Status
N1: r_d independence	α insensitive to r_d within Planck posterior	✓ Passed
N2: σ_8 prior	Signal strengthens under tight prior ($1.7\sigma \rightarrow 2.2\sigma$)	✓ Passed
T7: Leave-one-out	7/7 pass, $\alpha \in [-1.11, -0.88]$	✓ Passed

The $\sim 2\sigma$ preference for $\alpha < 0$ in the EFC growth sector is:

- Not driven by sound-horizon calibration choices.
- Not an artefact of prior width on σ_8 .
- Not carried by any single $f\sigma_8$ measurement.
- Distributed across the full redshift range $z = 0.02$ – 0.85 .

4.2 What this does not establish

This analysis does not constitute a detection. A 2.20σ signal corresponds to $p \approx 0.028$ (one-sided), well below the 5σ discovery threshold and below the 3σ evidence threshold. The result is a *robust hint*—the signal has survived all controls tested but remains sub-threshold.

Furthermore, $\Delta\text{BIC} = -0.91$ for the reference fit (all 7 points) shows that the Bayesian penalty for the extra parameter nearly cancels the χ^2 improvement. The data do not yet demand $\alpha \neq 0$; they are consistent with it.

4.3 Degeneracy structure

The positive correlation $r(\alpha, \Omega_m) \approx +0.59$ indicates a geometric degeneracy: higher Ω_m can partially absorb the effect of negative α . The anti-correlation $r(\alpha, \sigma_8) \approx -0.27$ confirms that $\alpha < 0$ (growth suppression) trades off against lower σ_8 . Breaking these degeneracies requires independent constraints—particularly from weak lensing shear (S_8) and ISW cross-correlation.

4.4 Implications for the EFC validation programme

This result enters the EFC Validation Ledger as a “Completed (robust hint)” in the Growth/Dynamics sector (EFC-D, L2 regime). It complements the existing BAO geometry tests and provides the first controlled robustness analysis of the growth-sector channel.

4.5 Next steps

1. **Survey-split test:** Remove entire surveys (not single points) to check for survey-level systematics.
2. **Extended growth data:** Incorporate DESI DR1 $f\sigma_8$ measurements when available ($\sim 20+$ points, sub-percent precision).
3. **Lensing cross-check:** Use KiDS/DES weak lensing to break the α – Ω_m degeneracy independently.
4. **ISW tomographic prediction:** The pre-registered EFC prediction ($\mathcal{L} > 0.7$, $A_{\text{ISW}} < 0.5$) provides the sharpest forthcoming kill test.

5. Reproducibility

All computations use the MVP-G1 pipeline within the EFC computational framework. Data sources are publicly available: BOSS DR12 consensus BAO [3], and the $f\sigma_8$ compilation from the references therein. The N2a nuisance configuration, LOO procedure, and pass criteria are fully specified in Section 2.

The analysis is tracked in the Symbiose knowledge system (Neo4j + Qdrant) with full provenance, and in the public EFC Validation Ledger at <https://energyflow-cosmology.com/validation-ledger/>.

Source code and configuration files are available in the EFC GitHub repository: <https://github.com/supertedai/EFC>.

References

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